

Analyzing the Effect the Concentration of Catalyst has on the Belousov-Zhabotinsky Reaction

How will the concentration of the ferroin at 0.001M, 0.0015M, 0.002M, 0.0025M, 0.003M affect the reaction rate of the Belousov-Zhabotinsky reaction by measuring the time difference between troughs in the absorbance wave?

Chemistry SL Internal Assessment

Candidate Number: lfg856

Research Question:

How will the concentration of the ferroin at 0.001M, 0.0015M, 0.002M, 0.0025M, 0.003M affect the reaction rate of the Belousov-Zhabotinsky reaction by measuring the time difference between troughs and crests on the absorbance wave?

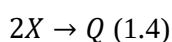
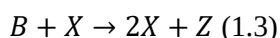
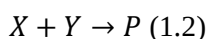
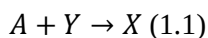
Introduction:

From the Golden Ratio to the Riemann Zeta function, the world has numerous amounts of patterns modeling both the beauty of symmetry and asymmetry in our world. This beauty can be modeled by chemical oscillating reactions, which display chaotic behavior ultimately conforming to a pattern that repeats itself. These reactions are central in our everyday lives, ranging from the circadian rhythm of humans to the drug delivery process in our bodies. These reactions are characterized by a periodic change of concentration in one or more components in the mixture. During these oscillating patterns, they exhibit reactions with far-from equilibrium behavior, showing that these reactions aren't dominated by equilibrium thermodynamic behavior. One of the most fundamental experiments exhibiting the behaviors of an oscillating reaction is the Belousov-Zhabotinsky (BZ). When plotted these reactions exhibit a wave-like pattern; a behavior that is used in everyday life ranging from soundwaves to analyzing cosmic bodies. The plot becomes especially useful for models that we can't physically manipulate such as the heart and its cardiac electrical system.

By learning about this behavior that underlies our world, I questioned how the reaction would change if the parameters were modified. The only necessity in the BZ-Reaction is the bromate oxidizer. In this experiment, I used sodium bromate (NaBrO_3) as the bromate oxidizer. To change the rate of this reaction, I decided to modify the concentration of the catalyst, ferroin, as many oscillating in nature rely on the concentration of the catalyst. For example, in the sleep cycle, the increase in melatonin (the catalyst) results in better sleep quality. This investigation will find how the concentration of catalysts affects the rate of reaction. This will be done by comparing BZ reactions with different concentrations of ferroin on the time between each color change. From this assessment, I hope to shed new light on how nature operates in its chaotic beauty and allow us to gain a better understanding of how this complex world works.

Background Information:

The BZ reaction exhibits a periodic change in the concentration of the reactants, intermediate species, and products. This change leads to a sequential pattern of the formation and disappearance of colored circles evolving chaotically, ultimately stopping when the reagents are consumed over the course of the reaction. The color changes demonstrated in the reaction results from the concentration of the intermediate species Br^- . An example of an oscillation pattern of a periodic oscillating reaction can be shown and characterized by the following set of equations:



“A and B are reactants; X, Y, and Z are intermediates; and P and Q are products” (Tuckerman).

Let k_1, k_2, k_3, k_4, k_5 denote the reactions rate of equations (1.1), (1.2), (1.3), (1.4), (1.5), respectively. We assume the concentrations of A and B to be large, so the $[A]$ and $[B]$ are constant over time. We can write rate equation of $[X]$ to be:

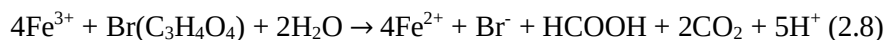
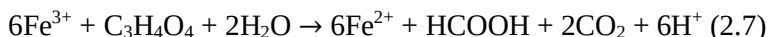
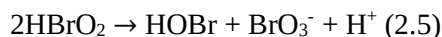
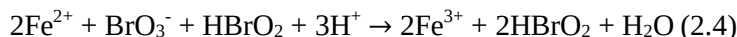
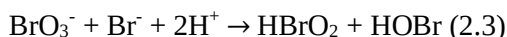
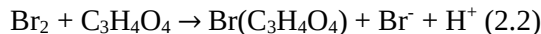
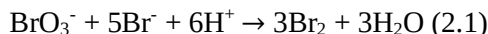
$$\frac{d[X]}{dt} = k_1[A][Y] - k_2[X][Y] + k_3[B][X] - 2k_4[X]^2$$

Because the intermediate fluctuates in respect to one another, i.e. when $[X]$ is large, $[Y]$ is small, and vice versa, we can either assume $[Y] \gg [X]$ or $[X] \gg [Y]$, which result in (1.1) and (1.2) or (1.3) and (1.4) being the deciding equations for the concentrations of $[X]$, respectively. Following this we can set $\frac{d[X]}{dt} = 0$ because $[X]$ and $[Y]$ reach a steady state as either $[X]$ or $[Y]$ is maximized while the other is minimized. Therefore, we can derive the following through steady-state approximation:

$$\text{When } [Y] \gg [X]: 0 \approx k_1[A][Y] - k_2[X][Y], \text{ leading to: } [X] \approx \frac{k_1[A]}{k_2}$$

$$\text{When } [X] \gg [Y]: 0 \approx k_3[B][X] - 2k_4[X]^2, \text{ leading to: } [X] \approx \frac{k_3[B]}{2k_4}$$

These two limits show that as $[X]$ will eventually reach an upper bound before decreasing to a lower bound, which will result in $[Y]$ reaching a lower bound and higher bound, respectively. More formally, the feedback is characterized when X reacts with B to form more X in (1.3), the increase in $[Y]$ after (1.5), and the consumption of $[X]$ in (1.2), (1.3) and $[Y]$ in (1.2). Similar to the above reactions, the BZ reaction used by this IA can be classified as follows:



The color change (red to blue and vice versa) in this reaction stems from equations (2.4), (2.7), and (2.8). In equation (2.4), the oxidation of the iron in the ferroin brings it from the red 2+ form to the blue 3+ form. In equation (2.7) and (2.8) the ferroin is reduced to its 2+ state from the malonic acid ($\text{C}_3\text{H}_4\text{O}_4$) and bromo-malonic acid ($\text{Br}(\text{C}_3\text{H}_4\text{O}_4)$).

The rate of this reaction can be measured by the change in concentration of reactants, products, or intermediate species per unit of time. Collision theory states that for any reaction to be successful, the particles must collide with sufficient energy (activation energy) and the proper orientation. The rate of any reaction can be influenced by many factors, such as temperature, pressure, surface area and catalysts. This IA will analyze the affect catalysts have on the rate of reaction.

A catalyst is “any substance that increases the rate of reaction without itself being consumed” (The Editors of Encyclopaedia Britannica). It does this through providing an alternate pathway with lower

activation energy. This increases the concentration of products because a greater number of reactants are able to overcome the lowered activation energy, allowing them to be products, which is seen in the Maxwell-Boltzmann distribution curve. The catalyst in this reaction is ferriox and increases the rate of reaction by forming temporary bonds with the reactants that are not permanent. These temporary bonds are broken by the molecules attached to the ferriox react with other molecules.

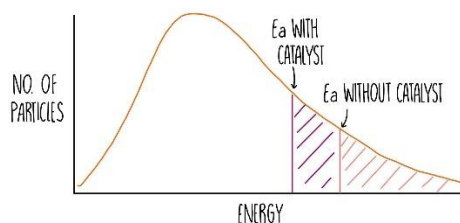


Figure 1: Maxwell-Boltzmann Distribution Curve showing the lower activation energy necessary for products to form. The increased area bounded by the curves shows the increased amount of particles that undergo the reaction (Reaction Rates).

The relation between activation energy and rate of reaction can be seen in the Arrhenius Equation:

$$k = Ae^{\frac{-E_a}{RT}}$$

“k denotes the rate constant of the reaction, A denotes the pre-exponential factor (the frequency of correctly oriented collisions between the reacting species), E_a denotes the activation energy in the reaction, R denotes the Universal Gas Constant, T denotes the absolute temperature associated with the reaction in Kelvin” (Byju’s).

Because temperature is held constant in this experiment, looking at the equation where E_a is the independent variable and k is the dependent variable, as E_a increases, k decreases, and vice versa. This means the rate of reaction decreases or speeds up, depending on the activation energy.

Hypothesis:

Given the information in the background, I predict that increasing the concentration of catalyst, ferrous sulfate, will increase the reaction rate of the BZ-reaction. This is because of the lower activation energy needed for reactions to occur, which leads to a decreased distance between trough and crest.

Variables:

Independent Variable:

Table 1: Independent Variable

Independent Variable	How it was Changed	Significance
Concentration of Catalyst (M)	The total solution amount (Reagent A & Reagent B & Reagent C & ferriox) is held constant, while the proportions of the reagents is changed, allowing for a difference in molarity of the ferrous sulfate.	The different concentrations of the catalyst will lead to a change in transparency of the solution. The period of the oscillations along with the peaks will be different demonstrating the affect catalysts have on the reaction.

Dependent Variable:

Table 2: Dependent Variable

Dependent Variable	How it Was Changed	Significance
Oscillation Period and Amplitude of the Colorimeter Readings	Changed as a result of the different concentration of catalyst	The change in period would call for the conclusion that the rate of reaction is increased if an increased concentration is added in the solution

Controlled Variables:

Table 3: Controlled Variables:

Controlled Variable	Effect if Not Controlled	How it Was Controlled
Concentration of Reagent A	The difference in concentration will change the rate of the reaction because it will cause an excess in some reactions but a dampening in others. To ensure this doesn't influence the results, it must be held constant	It was controlled through accurate measurement and weighing of the reactants prior to the reaction taking place.
Concentration of Reagent B		
Concentration of Reagent C		
Room Temperature	From the Arrhenius Equation, temperature affects the rate constant of the reaction; therefore, to ensure the only independent variable is the concentration of the catalyst an only thing affecting the reaction, temperature must be held constant	The experiment was done inside a classroom, where the temperature of the room was set to be constant.
Light Intensity	For consistent readings of the colorimeter, the light intensity must be the same when measuring transparency and absorbance.	The experiment was done inside a fume hood, the lighting of the room was set to be the same throughout the runs, and the wavelength of measurement inside the colorimeter was set to be the same
Equipment	The same equipment was used consistently. This prevented any incorrect readings that differed from previous runs. In addition, all the equipment was dried beforehand to ensure no residue was carried on from the previous run	The measuring devices that were used were dry, if not, it was dried with a paper towel. The same colorimeter and size of beakers were used across all experiments

Apparatus:

Lab Tools:

Table 4: Tool List

Item Name	Quantity	Uncertainty
2 mL Serological Graduated Pipette	3	± 0.01 mL
Controller Pipette	1	-
Colorimeter	1	-
Cuvettes	16	-
10 mL Graduated Cylinder	3	± 0.2 mL
Lens Paper	25	-
200 mL Graduated Cylinder	1	± 1 mL
300 mL Volumetric Cylinder	1	± 1 mL
150 mL Flask	3	± 1 mL
Pipette	1	-
Weigh Boats	3	-
Electric Scale	1	± 0.001 mL

Chemicals:

Table 5: Chemical List

Chemical Name	Quantity	Uncertainty
Concentrated Sulfuric Acid	18 mL	± 1.5 mL
Sodium Bromate	45 g	± 0.003 g
Malonic Acid	9 g	± 0.003 g
Sodium Bromide	9 g	± 0.003 g
Ferrous Sulfate	5.22 mL	± 1.65 mL
Distilled Water	783mL	± 3 mL

Risk Assessment:

Table 6: Risk Assessment

Chemical Name	Hazard	Pictogram	Preventive Measures
Concentrated Sulfuric Acid	Conc. Sulfuric Acid causes irritation/corrosion on the skin, serious eye damage/irritation, and organ toxicity and damage to the respiratory system, if ingested (Sulfuric Acid Safety Data Sheet)	 	The scientist wore goggles along with gloves and a lab coat to protect their skin when managing the chemicals in the solution. Furthermore, a hood was utilized to protect the scientist from the fumes generated when mixing the reagents and the solution.
Sodium Bromate	Sodium Bromate causes oral toxicity, organ toxicity and damage to respiratory system, if ingested, and skin corrosion/irritation, and serious eye damage/irritation (Sodium		

	Bromate Safety Data Sheet)		
Malonic Acid	Malonic Acid causes serious eye damage/irritation (Malonic Acid Safety Data Sheet)		
Sodium Bromide	Sodium Bromide is not considered hazardous	-	
Ferrous Sulfate	Ferrous Sulfate causes skin corrosion/irritation and serious eye damage/irritation, and oral toxicity, if ingested (Ferrous Sulfate Safety Data Sheet)		

Environmental Consideration: This experiment was done in a fume hood to prevent the buildup of harmful fumes generated from the reagents and the mixture of them particularly A and C. The improper disposal of conc. sulfuric acid, malonic acid, and ferrous sulfate may lead to harm and corrosion to the environment. Finn Disposal Guidelines and local regulations were used for disposing of the aforementioned chemicals. I used Flinn Disposal Methods 12a, 23, 24a, 24b, to properly dispose of my chemicals.

Ethical Concerns: This experiment poses no ethical concerns.

Exploring Methods of Procedures:

Initially to measure the rate of the reaction I planned to measure the time it took for the reaction to change color and measure the intervals with stopwatch; then, compare these times. However, I realized that timing the change of color with my own perception would lead to a tremendous amount of human error. To overcome this, I utilized a colorimeter to log the transparency difference that would arise from the solution changing color. However, I soon realized that the carbonation generated from the reaction, skewed the readings in absorbance and transparency, as it wasn't the pure solution. To prevent this, I started shaking the cuvette periodically to stop this buildup, however the carbonation was not entirely eliminated.

Procedure:

1. Prepare the solution for Reagents A, B, and C.
 - a. Reagent A: Mix together 201 mL of distilled water, 6mL of concentrated sulfuric acid, and 15g of sodium bromate.
 - b. Reagent B: Mix together 3g of malonic acid and 30 mL of distilled water.
 - c. Reagent C: Mix together 3g of sodium bromide with 30 mL of distilled water.
2. Pour out each reagent into their own respective flask to use for the solution.
3. Pour 4 mL of distilled water, using a 10mL graduated cylinder, into a cuvette that is used for calibrating the colorimeter, wipe possible leakage with lens paper.
4. For the 0.001M solution, mix together 3.07 mL of Reagent A (using a 10 mL graduated cylinder with a pipette), 0.51 mL of Reagent B (using a controller pipette and graduated pipette), 0.26 mL of Reagent C (using a controller pipette and graduated pipette), and 0.16 mL of Ferrous Sulfate into a cuvette (using a controller pipette and graduated pipette).

5. Shake the cuvette and clean the outside by wiping possible leakage with lens paper.
6. Calibrate the colorimeter using distilled water and set the transparency to 100% for distilled water.
7. Place the cuvette with the solution into the colorimeter and log the data for 10 minutes. Every interval of one minute, stop logging, take the cuvette out and shake it, then put the cuvette back in and resume logging.
8. Repeat steps 5-8, using different molarity solutions as listed below:
 - a. 0.0015M: 3mL of Reagent A, 0.5 mL of Reagent B, 0.25 mL of Reagent C, 0.25 mL of Ferrous Sulfate
 - b. 0.002M: 2.93mL of Reagent A, 0.49 mL of Reagent B, 0.25 mL of Reagent C, 0.34 mL of Ferrous Sulfate
 - c. 0.0025M: 2.85mL of Reagent A, 0.47 mL of Reagent B, 0.24 mL of Reagent C, 0.44 mL of Ferrous Sulfate
 - d. 0.003M: 2.76mL of Reagent A, 0.46 mL of Reagent B, 0.23 mL of Reagent C, 0.55 mL of Ferrous Sulfate
9. Repeat Steps 5-9 for Trial 2 and Trial 3
10. Pour the solutions from the cuvettes and remaining reagents into a tub filled with water to dilute for disposal.
11. Check the pH of the mixed liquids.
 - a. If the pH is between 6 and 8, it's safe for disposal.
 - b. Otherwise add baking soda and mix, until pH is at its desired state ($6 \leq \text{pH of mixed liquids} \leq 8$)
12. Pour the mixture into the drain for disposal.

Data Collection:

Qualitative Data:

1. Before, the mixture of Reagent of A and B were odorless and colorless. However, after Reagent C was added, an odor was released, and the mixture turned yellow.
2. Depending on the amount of Ferrous Sulfate added, the darker and thicker the solution was.
3. During the experiment, carbonation arose which caused the leakage of some of the cuvette.
4. It took a varied amount of time for each reaction to start (sometimes it never started).
5. When the cuvette was taken out, the solution inside seemed more liquidly, and sometimes was in a blue state compared to the initial darkish yellow state.
6. The cuvette was stained with the solution, even after rinsing and washing.

Raw Quantitative Data:

In the following data, an oscillation is defined where there is a measurable jump from trough to crest. In the following, peak is defined as a crest or a trough. The time for a reaction to start is defined as the time where the first peak occurs.

Table 7: 0.001M

Trials	Number of Oscillations	Time taken for reaction to start (± 1 seconds)	Maximum Absorbance (± 0.0005)	Minimum Absorbance (± 0.0005)	Maximum Transmittance (± 0.005 %T)	Minimum Transmittance (± 0.005 %T)
1*	2	165	1.115	-0.096	124.88	7.70

2	11	273	0.795	-0.125	133.21	16.01
3*	5	6	2.532	-0.068	116.95	0.29

Table 8: 0.0015M

Trials	Number of Oscillations	Time taken for reaction to start (± 1 seconds)	Maximum Absorbance (± 0.0005)	Minimum Absorbance (± 0.0005)	Maximum Transmittance (± 0.005 %T)	Minimum Transmittance (± 0.005 %T)
1	27	531	0.875	-0.113	129.81	13.34
2*	5	24	0.75	-0.129	134.73	17.78
3	6	597	2.348	-0.068	116.92	0.45

Table 9: 0.002M

Trials	Number of Oscillations	Time taken for reaction to start (± 1 seconds)	Maximum Absorbance (± 0.0005)	Minimum Absorbance (± 0.0005)	Maximum Transmittance (± 0.005 %T)	Minimum Transmittance (± 0.005 %T)
1	26	496	0.993	-0.105	127.34	10.16
2*	12	3	0.901	-0.110	128.94	12.56
3*	0	-	-0.035	-0.035	108.37	108.32

Table 10: 0.0025M

Trials	Number of Oscillations	Time taken for reaction to start (± 1 seconds)	Maximum Absorbance (± 0.0005)	Minimum Absorbance (± 0.0005)	Maximum Transmittance (± 0.005 %T)	Minimum Transmittance (± 0.005 %T)
1*	0	-	-0.151	-0.153	142.14	141.49
2	19	234 $\pm$ 1	0.971	-0.107	127.89	10.7
3*	2	676 $\pm$ 1	2.315	-0.069	117.11	0.48

Table 11: 0.003M

Trials	Number of Oscillations	Time taken for reaction to start (± 1 seconds)	Maximum Absorbance (± 0.0005)	Minimum Absorbance (± 0.0005)	Maximum Transmittance (± 0.005 %T)	Minimum Transmittance (± 0.005 %T)
1*	0	-	-0.164	-0.166	146.61	145.94
2	23	658	0.875	-0.116	130.48	13.34
3*	12	853	2.327	-0.071	117.75	117.75

(*) Trials marked with a “*” are deemed outliers due to the lack of oscillation behavior displayed in the time given. This was caused as a result of systematic and human error.

Data Processing/Analysis:

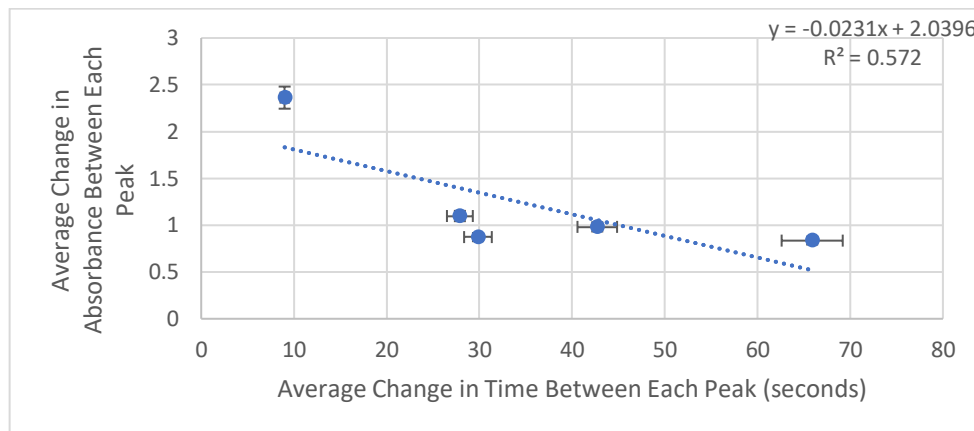
To calculate the distance between the peaks, the distance between each trough and crest was taken.

Table 12: Average Distance between Peaks in Seconds and Absorbance, Excluding Outliers

Molarity (M)	Average Change in Time Between Each Peak (± 1 seconds)	Average Change in Absorbance Between Each Peak (± 0.0005)
0.001	65.9	0.8352
0.0015	29.85	0.8727

0.002	27.9	1.0978
0.0025	42.72	0.9819
0.003	9	2.3628

Graph 1: Plot of Table 12, with the time between peaks as the x-axis and the change in absorbance between peaks as the y-axis



Discussion of Outliers:

Through the analysis of the graphs in Appendixes 1-15, it is apparent that many of the graphs did not match the patterns traditionally shown in the BZ Reaction – color change and frequency of oscillations. Such outliers are left out of the data analysis to prevent them from affecting the data. The outliers may be caused by random error from equipment, and human error when measuring the reactants for the reagents and the ratio of the reagents in the solution. The error in measuring the reactants impacts the reactions of the BZ-reaction, possibly decreasing the rate of reaction due to the lack of collisions that occur. This would explain the flat line result that occurred in some of the data logging (as shown in Appendix 1). In addition, human error from measuring the ratio of the reagents in the final solution may also cause the outliers. The incorrect ratio may have led to the system of reactions from 2.1-2.3 of the BZ reaction to overpower the generation of Fe^{3+} leading to a solution without the full color change, explaining situations shown in Appendix 5. Lastly the readings of the colorimeter may affect the data logged for higher molarity, as it measures the absorbance and transparency in increments of three seconds. However, I find this possibility to be unlikely because it is highly improbable that over the course of 10 minutes every reaction peak occurs during a time where the colorimeter does not log it.

Uncertainty Calculations:

Because the uncertainty of the scale is 0.001g, 0.01mL for the 2mL graduated pipette, 0.2 mL for the 10mL graduated cylinder, and 1mL for the 150mL flask, 200mL cylinder, and 300 mL cylinder, the following uncertainty calculations follow for each calculation:

Measuring out the reagents:

$$\left(\left(\frac{1}{201} + \frac{0.2}{6} + \frac{0.001}{15} \right) + \left(\frac{0.001}{3} + \frac{1}{30} \right) + \left(\frac{0.001}{3} + \frac{1}{30} \right) \right) * 100 = 11\% \text{ after significant figures}$$

Measuring the amount for each run:

$$0.001\text{M}: \left(\frac{0.2}{3.07} + \frac{0.01}{0.51} + \frac{0.01}{0.26} + \frac{0.01}{0.16} \right) * 100 = 19\% \text{ after significant figures}$$

$$0.0015\text{M}: \left(\frac{0.2}{3} + \frac{0.01}{0.5} + \frac{0.01}{0.25} + \frac{0.01}{0.25} \right) * 100 = 17\% \text{ after significant figures}$$

$$0.002\text{M}: \left(\frac{0.2}{2.93} + \frac{0.01}{0.49} + \frac{0.01}{0.25} + \frac{0.01}{0.34} \right) * 100 = 16\% \text{ after significant figures}$$

$$0.0025\text{M}: \left(\frac{0.2}{2.85} + \frac{0.01}{0.47} + \frac{0.01}{0.24} + \frac{0.01}{0.44} \right) * 100 = 16\% \text{ after significant figures}$$

$$0.003\text{M}: \left(\frac{0.2}{2.76} + \frac{0.01}{0.46} + \frac{0.01}{0.23} + \frac{0.01}{0.55} \right) * 100 = 16\% \text{ after significant figures}$$

The average percentage error over these runs is 16.8%. Therefore, the total average percent of uncertainty of the reaction is 27.8%.

Statistical Analysis:

The data will be evaluated using the Pearson Correlation Coefficient, given by the following formula:

$$r = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}}$$

$$r = \frac{(65.9 - 35.074)(0.8352 - 1.246764) + (29.85 - 35.074)(0.8727 - 1.246764) + (27.9 - 35.074)(1.0978 - 1.246764) + (42.72 - 35.074)(0.9819 - 1.246764) + (9 - 35.074)(2.3628 - 1.246764)}{\sqrt{((65.9 - 35.074)^2 + (29.85 - 35.074)^2 + (27.9 - 35.074)^2 + (42.72 - 35.074)^2 + (9 - 35.074)^2)((0.8352 - 1.246764)^2 + (0.8727 - 1.246764)^2 + (1.0978 - 1.246764)^2 + (0.9819 - 1.246764)^2 + (2.3628 - 1.246764)^2)}}$$

From this calculation, a Pearson Correlation Coefficient of $r = -0.7563$ is derived.

Analysis of Data:

Through analyzing the data and excluding the outliers, there seemed to be a decreased time of oscillation and higher change in absorbance as the molarity increases. Furthermore, the Correlation Coefficient of -0.7563 demonstrates the strong correlation between the concentration of catalyst and the time taken for each oscillation. Due to the relatively high uncertainty of 27.8%, systematic and methodological error may have caused the excess number of outliers.

Evaluation:

Conclusion:

In conclusion, this investigation aimed to explore how different concentrations of ferriox-sulfate (0.001M, 0.0015M, 0.002M, 0.0025M, 0.003M) will affect the rate of reaction of the BZ-reaction. Through the collected data, it is difficult to determine a finalized answer due to the numerous outliers in the dataset. If the outliers are excluded, the data seems to show that as the concentration of catalyst increases the rate of reaction also increases, as demonstrated through the Pearson Correlation Coefficient being -0.7563. However, an excess in catalyst may lead to minimal change in the rate of reaction and may instead hurt the reaction, as seen with the 0.0025M and 0.003M solutions. This excess does nothing, as all the reactants are already paired with a catalyst to lower the activation energy, so having an excess in catalyst does nothing. Furthermore, because all the reactants have a lower activation energy, they collide more frequently to form more products and vice versa. This increase in products and reactants causes them to take space on the surface of the catalyst, as shown in equations 2.4, 2.7, and 2.8. The space that is taken up by the reactants and products waste reactor space because there is nothing for them to react with (Clark). In addition, the extra reactants and products that are generated may lead to side reactions or undesired byproducts affecting the balance of the equation's proportions ('Negative'). Therefore, the excess may slow the generation of molecules, such as bromide and bromine, thus slowing the reaction as a whole. I also noticed that the increased catalyst increases the amplitude of the graph given by the

colorimeter, meaning an increased change in color between the two states of the reaction. This is shown most prominently in Appendix 16, where it has the lowest amplitude, while having the lowest concentration: 0.0005M. To conclude, because many of my experiments resulted in outliers that deviated from the convention of the BZ reaction, I cannot conclude that my hypothesis is correct or incorrect. However, I am confident that the correlation between the concentration of the catalyst and rate of reaction exists and can be proven through more trials and more intensive control. This correlation arises as “A larger proportion of the collisions that occur between reactants have enough energy to overcome the activation energy for the reaction. As a result, the rate of reaction increases” (Bodner Research).

Strengths and Weaknesses:

a. Strengths:

1. The visualization of the data logging equipment, which provided an analytical way to assess, how an oscillation changed, reducing the human error associated determining what constitutes a cycle.
2. The 3 trials conducted allowed for more data, and produced a fair and credible test, to check the data with the convention of an oscillating reaction.

b. Limitations and Weakness:

1. It's possible that the colorimeter may have missed some peaks as the colorimeter measures in increments of three seconds.
2. Because of the size of the cuvette, I may have mismeasured or misread the instructed measurements, as shown with the relatively high total percent uncertainty.

Table 13: Weakness and Improvement

Weakness	Impact of Weakness	Suggested Improvement
Fluidity of the Mixture (Systematic Error, Methodological)	When the mixture wasn't properly mixed the liquids with the higher density float to the top, while the liquids with the lower density floated to the bottom, leading to the solutions not being able to be mixed properly. This may have led to false data and high inaccuracies as shown by the outlier. The dried ferrous sulfate on the rim of the bottle, may have gotten into the cuvette's, causing it to remain at the bottom	Before pouring the ferrous sulfate out, clean the beaker, to avoid any previous residue, along with the rim of the ferrous sulfate bottle. Furthermore, during a trial, I could take the cuvette out and shake it more often, ensuring that the mixture is mixed.
Generation of Carbonation (Systematic Error, Methodological)	The generation of carbonation caused bubbles to arise in the cuvette. These bubbles may have blocked the light absorbance and intensity of the colorimeter, leading to false data. Furthermore, the carbonation bubbles, caused overflowing issues in the cuvette spilling some of the solution inside the	Instead of filling the cuvette to its brim, I could leave some room, so even if the carbon bubbles arise, the solution wouldn't leak. In addition, I could take the lid of the cuvette in increments, to prevent the build of carbonation, and ensure it's at the minimum.

	colorimeter, affecting the results, due the blockage of the light.	
Measurement of Reagents in the Cuvette (Random Error)	Due to the small volume of the cuvette and number of reagents as mentioned in the previous section, the ratio of the solution may have been different from the intended ratio. This imbalance would have caused an inconsistency in how the reactions would have occurred, offsetting the reactants, intermediates, and products in the reaction.	I would change the device chosen to measure the reaction, utilizing a lux sensor instead of the colorimeter. Previously I thought that the colorimeter would be a more accurate option due to the closed nature of its system – trapping a cuvette in darkness, and only having waves pass through. If the lux sensor is utilized, I could also place the mixture on an automated mixer due to the increased volume, also preventing the aforementioned weaknesses.

Extensions:

A future investigation, building off this experiment, may investigate another property of the BZ-reaction: spatial patterns. When this reaction is examined in a petri dish, complex and intricate spirals form as a result of the mixture of the reagents. The understanding and analysis may lead to notable contributions in biological dynamics, where the oscillations and patterns of this system are similar to natural biological phenomena. For example, cardiac arrhythmias exhibit similar wave propagation, and allows the heart to be studied through the BZ-reaction rather than on the heart directly. In conclusion, the successful analysis of the BZ-reaction may lead to something that saves humanity.

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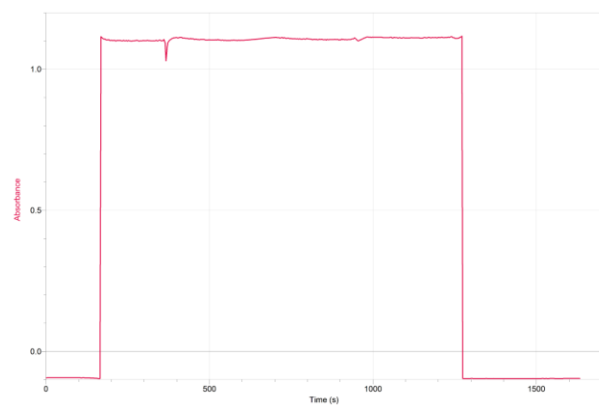
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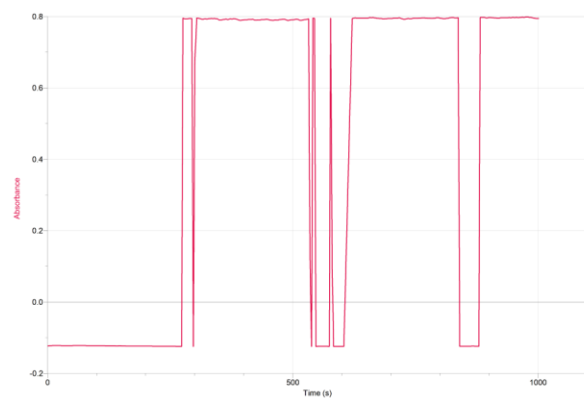
Appendix:

Appendix 1-3: Graphs of the Reaction for 0.001M:

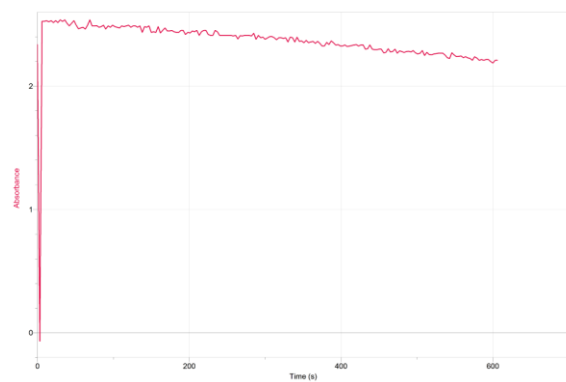
Appendix 1: Trial 1



Appendix 2: Trial 2

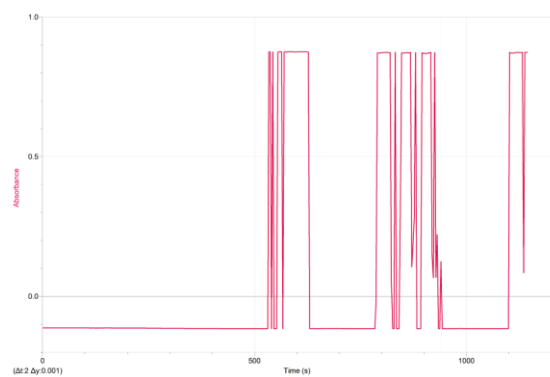


Appendix 3: Trial 3

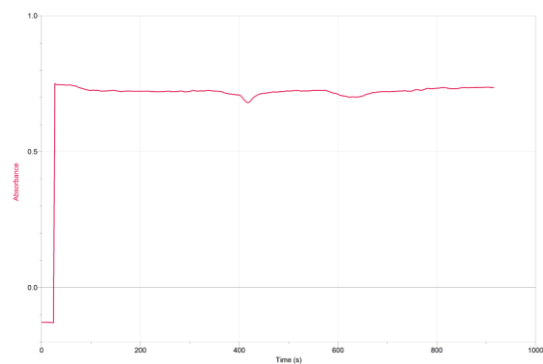


Appendix 4-6: Graphs of the Reaction for 0.0015M

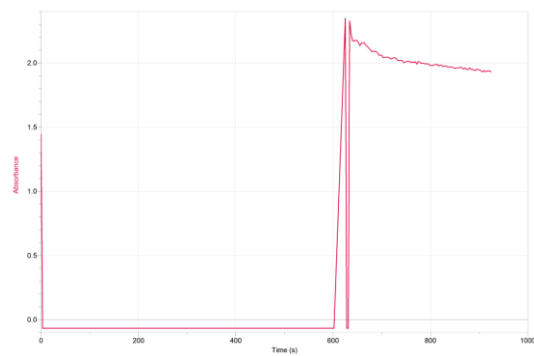
Appendix 4: Trial 1



Appendix 5: Trial 2

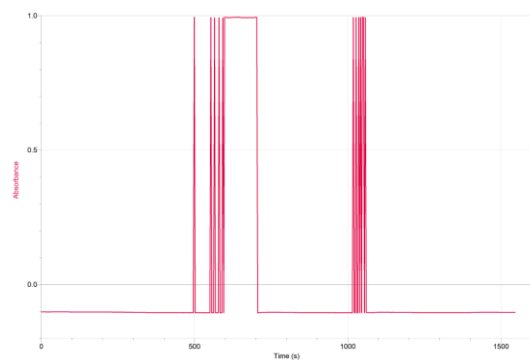


Appendix 6: Trial 3

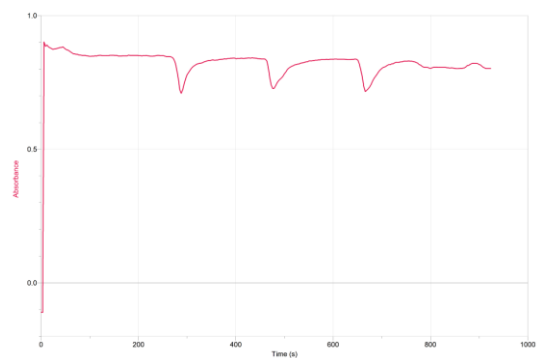


Appendix 7-9: Graphs of the Reaction for 0.002M

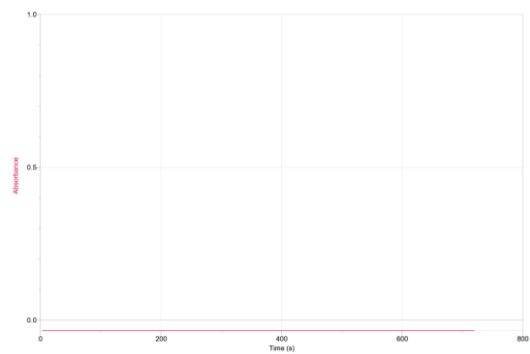
Appendix 7: Trial 1



Appendix 8: Trial 2

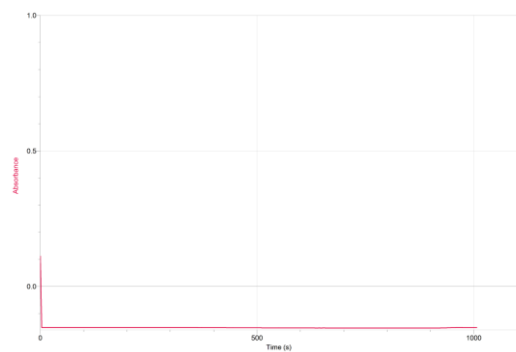


Appendix 9: Trial 3

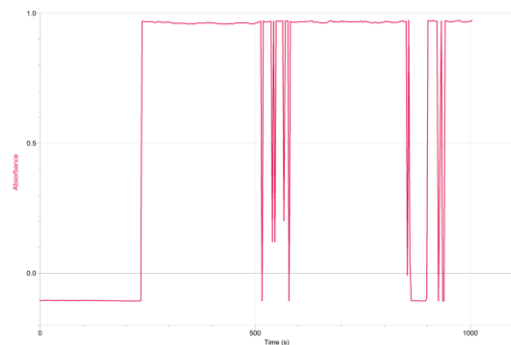


Appendix 10-12: Graphs of the Reaction for 0.0025M

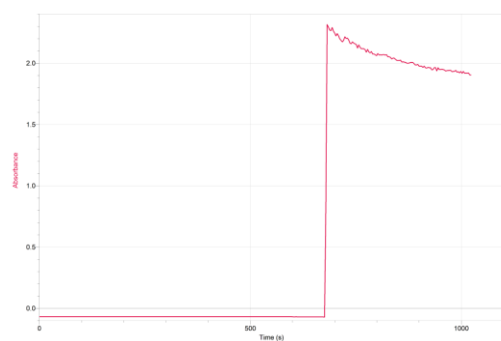
Appendix 10: Trial 1



Appendix 11: Trial 2

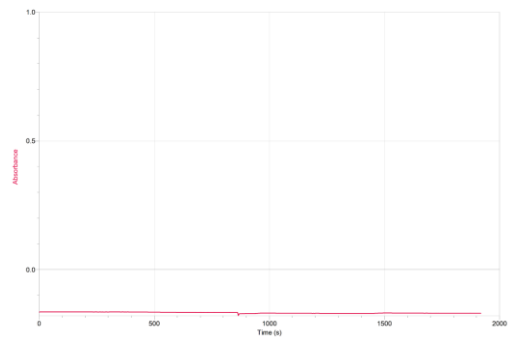


Appendix 12: Trial 3

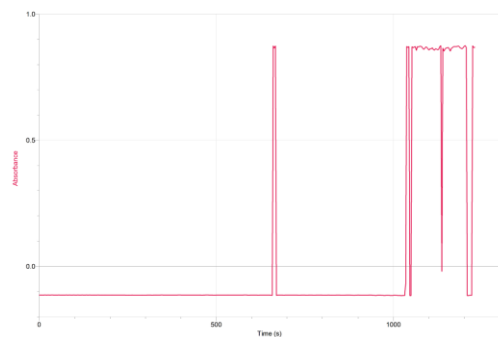


Appendix 13-15: Graphs of the Reaction for 0.003M

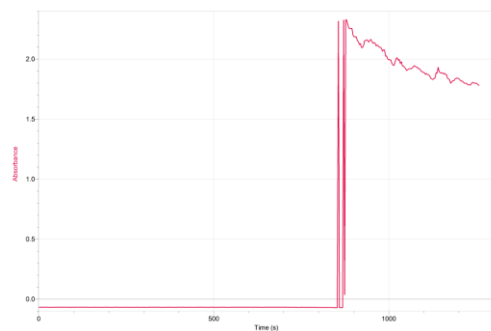
Appendix 13: Trial 1



Appendix 14: Trial 2



Appendix 15: Trial 3



Appendix 16: 0.0005M Trial

